

SYSTEMS PROJECT MANAGEMENT

Systems Analysis & Design

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Outline

- 1 Information Systems
- 2 Enterprises as Systems
- 3 Software Methodologies



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Information Systems

- An **Information System** is a **system** that *collects, processes, stores, and disseminates* information.
- Information systems are used to support and manage business operations.
- Information systems are used to automate and optimize business processes.
- Examples of information systems include transaction processing systems, management information systems, decision support systems, executive information systems, expert systems, and data systems.



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Data Systems

- A **Data System** is a **system** that *collects, processes, stores, and retrieves data*.
- **Data systems** are used to **store** and **analyze** data.
- Examples of **data systems** include databases, data warehouses, data lakes, data marts, data cubes, and data streams.



Expert Systems

- An **Expert System** is a system that uses **knowledge** and **reasoning** to solve problems.
- **Expert systems** are used to **automate** and **optimize** decision-making processes.
- Examples of **expert systems** include diagnostic systems, predictive systems, prescriptive systems, decision support systems, and automated reasoning systems.



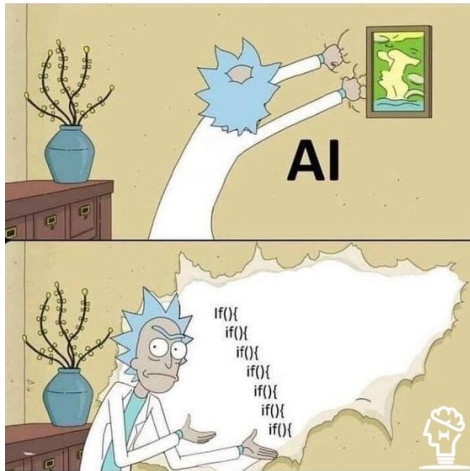
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Expert Systems as Classical Artificial Intelligence

Here is a great example of a **diagnostic system**.



Risks and Failures in Information

- **Information systems** are subject to **risks** and **failures** that can impact business operations.
- **Risks and failures** can be mitigated through **security measures**, **backup systems**, **disaster recovery plans**, and **monitoring tools**.
- Examples of **risks and failures** include security breaches, data loss, system downtime, performance issues, and compliance violations.



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Enterprises: Bottom-Up and Top-Down Approaches

- **Bottom-Up Approach:** Analyzes an enterprise by examining its individual units or components, then aggregating them to *understand* the entire organization.
- **Top-Down Approach:** Starts with an overall vision or strategy and decomposes it into subsystems, departments, and processes.



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PIECE Framework for Enterprises

- **Participation:** *Engaging stakeholders* at every level.
- **Independence of Thought:** Encouraging *diverse* and *innovative ideas*.
- **Elaboration:** Developing and *refining ideas* and *processes*.
- **Communication:** Ensuring clear and *effective exchange* of information.
- **Exploration:** Embracing *continuous innovation* and *improvement*.



Enterprise System Typologies

- **Rational Systems:** Organizations **driven by logical** and structured processes with clear hierarchies.
- **Natural Systems:** Organizations viewed as **organic entities** with informal structures and emergent behavior.
- **Open Systems:** Enterprises that continuously **interact** with their **external environment** for information, resources, and feedback.



Business Systems and Models

- **Business Systems:** Integrated frameworks that encompass an enterprise's processes, operations, and strategies.
- **Examples:** Enterprise Resource Planning (ERP) systems, Customer Relationship Management (CRM) systems, Supply Chain Management (SCM) systems.
- **Business Models:** Describe how an organization creates, delivers, and captures value.
 - Examples include subscription-based, freemium, platform-based, and direct sales models.



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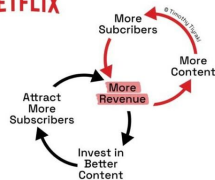
Business Models Examples

Understanding Business Models Through Flywheels

amazon



NETFLIX



Spotify



LinkedIn



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Basic Concepts in Software Methodologies

- **Software methodologies** provide frameworks for **planning**, **designing**, **developing**, **testing**, and **maintaining** software projects.
- They help teams manage project **complexity** and ensure quality deliverables.



Traditional Methodologies

- **Waterfall:** A linear sequential approach where each phase must be *completed before moving* to the next.
- Suitable for projects with well-defined requirements and *minimal uncertainty*.
- Emphasizes thorough documentation and up-front planning.



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Agile Methodologies

- Emphasizes **iterative development**, **customer collaboration**, and **adaptability to change**.
- Based on the **Agile Manifesto** (2001), which values **individuals and interactions** over processes and tools.
- *Examples* include Scrum, Kanban, Extreme Programming (XP), and Lean Software Development.
- **Agile methodologies** are suitable for projects with **evolving requirements** and **high uncertainty**.
- Promote **adaptive planning**, **evolutionary development**, and **early delivery** of working software.



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Case Study: Scrum

- **Scrum** employs short, time-boxed iterations called **sprints** (typically 1–4 weeks).
- Key events include **daily stand-ups**, **sprint planning**, **sprint reviews**, and **sprint retrospectives**.
- **Scrum** focuses on **adaptability** and **continuous improvement**.



Case Study: Kanban

- **Kanban** visualizes **work flow** on boards and limits Work In Progress (WIP) to optimize flow.
- Emphasizes **continuous improvement**, **flow optimization**, and **just-in-time delivery**.
- Ideal for **support teams** and **maintenance projects** requiring *continuous flow* with minimal iteration planning.



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

